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Quadrilaterals

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Quadrilaterals

A **polygon** is a closed, plane (flat) figure made up of line segments. In this case, *closed* means that the sides meet.



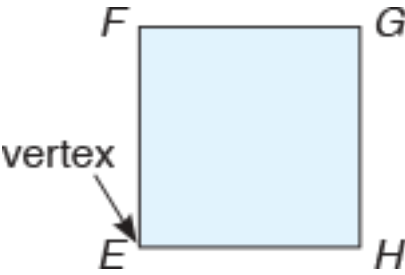
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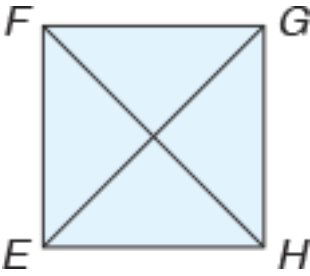
A **quadrilateral** is a polygon with four sides. Squares and rectangles are the two most common quadrilaterals. If you add the angles inside any quadrilateral, the sum is always 360° .

Look at the quadrilateral $EFGH$ shown here. Each time two sides meet, an angle is formed. The point where the sides meet is called a vertex. The four labeled points are the **vertices** of this figure.



Sides EF and GH are **opposite** (across from) each other. Sides FG and EH are also opposites. Sides EF and EH are **adjacent**. This means they share an endpoint.

A straight line connecting opposite vertices (corners) is called a **diagonal**. Line segment EG is a diagonal; so is FH .



Below are examples and properties of five common quadrilaterals.

Name	Properties	Examples
square	Opposite sides are parallel. All sides are equal. All angles are right angles (90°). Diagonals are equal and perpendicular.	
rectangle	Opposite sides are equal and parallel. All angles are right angles (90°). Diagonals are equal but not necessarily perpendicular.	
rhombus	Opposite sides are equal and parallel. All sides are equal. Diagonals are perpendicular but not necessarily equal.	
parallelogram	Opposite sides are equal and parallel.	
trapezoid	Only one pair of sides is parallel.	

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Lesson 1: Basic Figures

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